Lab 4: Simplified Markdown Version

Lab 4: Regression and Classification Evaluation Metrics

Lab 4: Regression and Classification Evaluation Metrics Part 1 — KNN Classification on the Breast Cancer Wisconsin (Diagnostic) Dataset, and Comparison with Regression Evaluation Metrics (Lab 3) Aim: To implement KNN classification on the Breast Cancer dataset and analyse model performance using train test split, heuristic K selection, cross validation, ROC AUC, and classification metrics.
Also, to compare classification metrics with the regression metrics studied in Linear Regression (Lab 3 — Simple Linear Regression: GPA Analysis and Multiple Linear Regression: Employee Salary Analysis).

Step 0 — Import Libraries

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Task 1: Data Preparation 1.1 — Load the dataset using sklearn

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Note on target encoding: in this scikit learn build of the dataset, target = 0 corresponds to malignant and target = 1 corresponds to...

1.2 — Convert into a DataFrame and explore structure

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Observation: All 30 features are continuous numeric measurements derived from digitised images of a breast mass (radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, fractal dimension — each reported as a mean , standard error , and "worst"/largest value).
The features vary hugely in scale — e.g.

1.3 — Check missing values and duplicates

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Observation: The dataset is clean out of the box — scikit learn's bundled copy has no missing values and no duplicate rows , so no imputation or de duplication is required (unlike the raw survey data in Lab 3, this dataset has already been curated by UCI).
The classes are moderately imbalanced (~63% benign vs ~37% malignant), which is an important point we return to in Task 5/6 — it is one reason accuracy alone is not a reliable metric for this problem.

1.4 — Feature scaling using StandardScaler

1.4 — Feature scaling using StandardScaler Why scaling matters for KNN: KNN classifies a point based on the (typically Euclidean) distance to its nearest neighbours.
If features are left on their raw scales, a feature like mean area (values in the hundreds to low thousands) will completely dominate the distance calculation over a feature like mean smoothness (values around 0.05–0.15), even if the smaller scale feature is just as informative for distinguishing malignant from benign tumours.

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Task 2: Train Test Split Analysis We split the (scaled) dataset into training and testing sets using three ratios — 80:20, 70:30, and 90:10...

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Observation: All three split ratios give a very similar accuracy (within roughly 1–2 percentage points of each other) because the dataset is reasonably sized (569 rows) and stratify=y keeps the malignant:benign class ratio consistent across every split.
The 90:10 split has the smallest test set, so its accuracy estimate has the highest variance (each misclassified point swings accuracy by a larger amount, e.g.

Task 3: KNN Model with Heuristic K Selection 3.1 — Heuristic method for K selection A commonly used heuristic for an initial guess at K is: $$K \approx \sqrt{n}$$ where n is the number of training samples.
The square root rule is a rough rule of thumb (not a theoretically optimal value) that tends to avoid K values that are either too small (noisy, overfit) or too large (oversmoothed) for typical dataset sizes.

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3.2 — Model training: baseline heuristic K, and K ± 5 sweep

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Observation: Accuracy stays within a fairly narrow band across this range of K, which is typical once K is already in a sensible neighbourhood — small changes in K rarely cause dramatic accuracy swings on a well scaled, well separated dataset like this one.
The best performing K in the local ±5 window is identified above and is compared against the cross validation result in Task 4 before making a final choice.

3.3 — Distance metrics, and decision boundary mapping

3.3 — Distance metrics, and decision boundary mapping Euclidean Distance $$d(p, q) = \sqrt{\sum_{i=1}^{n} (p_i q_i)^2}$$ This is the ordinary straight line ("as the crow flies") distance between two points in feature space.
It is the default and most common metric for KNN and works well when features are continuous, on comparable (scaled) ranges, and the geometric notion of "closeness" is meaningful — exactly the situation here after StandardScaler .

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Observation — how the boundary changes with K: K = 1 : the boundary is highly jagged / irregular , hugging individual training points closely.
This is the classic signature of overfitting / low bias, high variance — the model memorises noise and outliers in the training data.

Task 4: Cross Validation A single train test split gives only one estimate of accuracy, which depends on which rows happened to land in the test set.
K Fold Cross Validation instead splits the (training) data into k folds, trains on k–1 folds and validates on the remaining fold, rotating through all folds — giving k accuracy estimates whose mean is a far more reliable measure of how the model will generalise.

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Observation: Cross validation accuracies are smoother and more stable across K than the single train test split curve from Task 3 — because each point on the CV curve is already an average over 5 or 10 different train/validation partitions, whereas the Task 3 curve came from one fixed test set. The 5 fold and 10 fold curves broadly agree with each other and both sit close to the heuristic K, confirming that the √n rule gave a reasonable starting point here.

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Task 5: Classification Evaluation We now train the final KNN model (using the K selected in Task 4, on the standard 80:20 split from...

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Observation: The final KNN model (K = selected via cross validation) achieves high accuracy, precision, recall, and F1, along with an ROC AUC close to 1, indicating it separates the two classes very well on this dataset.
The confusion matrix additionally shows exactly which errors occur — in particular, any false positives (a malignant tumour classified as benign) are the clinically dangerous error type, since a missed cancer diagnosis is far costlier than a false alarm that gets caught on further testing.

Task 6: Comparative Study with Regression (Lab 3 Integration) Reference — Lab 3 (Linear Regression) results: | Experiment | R^2 | MAE | MSE...

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Regression vs Classification evaluation — conceptual differences

Regression vs Classification evaluation — conceptual differences Error based evaluation (Regression): the target is a continuous number (GPA, salary).
Metrics like MAE, MSE, RMSE, and R^2 all measure how far a numeric prediction is from the true value — they quantify a magnitude of error on the same (or a squared) scale as the target.

Task 7: Analytical Questions 1.
Why is KNN called a lazy learning algorithm?

Quick One Line Interpretations A condensed, at a glance summary of every key result in this lab. Data prep No missing values, no duplicates — dataset is clean and ready to use as is.

Conclusion Optimal K: the √n heuristic gave an initial estimate that was then refined using a local K ± 5 sweep on the train test split and confirmed against 5 fold and 10 fold cross validation; the cross validated result was adopted as the final K, since CV gives a more statistically reliable estimate than a single split.
Effect of train test split variations: the 80:20, 70:30, and 90:10 splits produced very similar accuracy for this dataset, but smaller test sets (90:10) give noisier, less stable accuracy estimates, while larger test sets (70:30) trade away some training data for a more statistically reliable evaluation — a direct illustration of the evaluation side bias–variance trade off discussed in Task 2.